Enes Arda

PhD student in Electrical and Computer Engineering, The Ohio State University

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Research interests

Machine learning theory and optimization, with a focus on reinforcement learning, Transformers, and state-space models. Earlier work in information theory and LLM-based semantic communication.

Education

The Ohio State University Aug 2025–May 2029 (expected)
PhD in Electrical and Computer Engineering. Advisor: Prof. Atilla Eryilmaz. GPA: 4.0/4.0.

The Ohio State University Aug 2023–May 2025
MS in Electrical and Computer Engineering. Advisor: Prof. Aylin Yener. GPA: 4.0/4.0.

Bogazici University Sep 2017–Nov 2022
BS in Electrical and Electronics Engineering, High Honors. GPA: 3.9/4.0.

Visiting studies: TU Darmstadt (Erasmus exchange, 2021); RWTH Aachen University (summer school, 2019).

Research experience

Graduate Research Assistant, OSU — Advisor: Prof. Atilla Eryilmaz Aug 2025–present

- Developed *belief geometry*, an analytical framework that compares attention and state-space models by abstracting capabilities common to many sequential learning problems. Derived sharp regret, memory, and width tradeoffs in controlled in-context-learning tasks, and showed experimentally with LLaMA-type Transformers and Mamba-2 that the architectural lessons extend beyond those settings (**preprint**).
- Established the first known linear convergence guarantee to the optimum, up to critic error, for policy mirror descent with linear function approximation in ergodic average-reward MDPs. Improved the mixing-time dependence of sample complexity from t_{mix}^5 to t_{mix}^3, with matching critic-estimation lower bounds in mixing time and accuracy (**NeurIPS 2026**).
- Proved finite-time training guarantees for shallow Transformers in the NTK regime: only logarithmic width dependence on the sample size and, thanks to softmax, no sequence-length dependence. By contrast, recurrent-network bounds can grow exponentially with sequence length (**AISTATS 2026**).

Graduate Research Assistant, INSPIRE Lab, OSU — Advisor: Prof. Aylin Yener Aug 2023–May 2025

- Developed a rate-distortion-perception framework that characterizes fundamental limits of text summarization and provides a practical benchmark for evaluating summarizers (**ISIT 2025**; **Allerton 2025**; IEEE Transactions on Communications, revision under review).
- Designed LLM-based semantic forwarding for cooperative relay networks, improving semantic fidelity over baselines under noisy channel conditions (**CISS 2024**; **ICC 2025**; **JSTSP**; **U.S. patent application**).

Professional experience

Model Research Scientist Intern, AllyIn.ai, Santa Clara, CA (Remote) Jun–Aug 2025
Built an NLP pipeline that mined patent filings to identify new products and estimate launch timing.
Built a sentiment-analysis pipeline to flag unusual earnings-call Q&A pairs for deeper LLM analysis.

Data Scientist, Spryfox GmbH, Darmstadt, Germany Jun 2022–Aug 2023
Built and deployed predictive models for pet-insurance applications, including disease-risk prediction.
Built a language-model pipeline to classify customer requests.
Designed bidding strategies to acquire homeowner project leads in competitive home-services auctions.

Part-time Machine Learning Engineer, Eatron Technologies, Istanbul, Turkey Feb–Jun 2022
Contributed to data processing and model training for electric-vehicle battery state-of-health prediction.

Publications and patents

1. **Enes Arda** and Atilla Eryilmaz. "Sharp Convergence and Sample Complexity of Policy Mirror Descent for Average-Reward MDPs." *NeurIPS 2026* (Accepted). arXiv:2610.04117 · Code
2. **Enes Arda**, Semih Cayci, and Atilla Eryilmaz. "A Comparative Analysis of Attention versus State-Space Models for In-Context Learning." *Under review*, 2026. arXiv:2609.32341 · Code
3. **Enes Arda**, Semih Cayci, and Atilla Eryilmaz. "Finite-Time Analysis of Gradient Descent for Shallow Transformers." *AISTATS 2026*, PMLR 300:3988–3996. arXiv:2601.16514 · PMLR · Code
4. **Enes Arda**, Emrecan Kutay, and Aylin Yener. "Semantic Relaying for Next Generation Networks: Harnessing Advances on Large Language Models." *IEEE Journal of Selected Topics in Signal Processing*, 2026. IEEE Xplore
5. **Enes Arda** and Aylin Yener. "Fundamental Limits for Text Summarization: A Rate-Distortion-Perception Approach." Revised manuscript under review at *IEEE Transactions on Communications*.
6. **Enes Arda** and Aylin Yener. "Rate-Distortion-Perception Trade-off in Summarization." *Allerton Conference on Communication, Control, and Computing*, 2025. Illinois repository
7. **Enes Arda** and Aylin Yener. "A Rate-Distortion Framework for Summarization." *IEEE International Symposium on Information Theory (ISIT)*, 2025. arXiv:2501.13100 · IEEE Xplore
8. **Enes Arda**, Emrecan Kutay, and Aylin Yener. "Semantic Relaying of Sentences." *IEEE International Conference on Communications (ICC)*, 2025. IEEE Xplore
9. **Enes Arda**, Emrecan Kutay, and Aylin Yener. "Semantic Forwarding for Next Generation Relay Networks." *Conference on Information Sciences and Systems (CISS)*, 2024. arXiv:2401.17247 · IEEE Xplore
10. **Enes Arda**, Emrecan Kutay, and Aylin Yener. "Cooperative Network for Semantic Communication." *U.S. patent application*, US 2026/0170248 A1, 2026.

Technical skills

Python, PyTorch (Lightning, FSDP), vLLM, Hugging Face Transformers, NumPy, SciPy, scikit-learn, Slurm-managed GPU clusters. Prior experience with Docker and AWS.

Languages: English (fluent), Turkish (native), German (limited working proficiency).

Honors and awards

- Dr. Burn Lin Travel Grant Award, The Ohio State University, 2026.
- High Honors, Bogazici University, 2022.
- Erasmus Exchange Program Grant, European Union, 2021.
- Merit Scholarship, Turkish Automobile Association, 2019–2022.
- DAAD Scholarship, RWTH Aachen University summer school, 2019.
- Ranked 389th out of 2.2 million candidates in Turkey's university entrance exam, 2017.

Academic service

Conference reviewer: *International Conference on Learning Representations* (ICLR 2026), *IEEE International Symposium on Information Theory* (ISIT 2024), *IEEE Information Theory Workshop* (ITW 2024).
Journal reviewer: *IEEE Transactions on Information Theory*, *IEEE Journal of Selected Topics in Signal Processing*, *IEEE Journal on Selected Areas in Communications*.

Mentoring and leadership

- Junior Student Mentor, Bogazici University Student Representative Board, 2019–2020.
- C and Arduino tutor for high-school students, 2018–2019.
- Chair, Robotics and Automation Society, Bogazici University IEEE Student Branch, 2018–2019.